

**Aeronive** Labs

WEBSITE DOCUMENTATION

Complete content & design record

Every page, section, headline, and body string on aeronivelabs.com, alongside the design system, the 3D architecture, and the technical decisions behind them.

Document	Scope	Stack	Version
Website content inventory	4 pages · 1 API route	Next.js 16 · React 19	1.0

Contents

1. **Overview** — what the site is and how it is put together
2. **Brand & positioning** — name, tagline, proposition, voice
3. **Design system** — colour, type, surfaces, motion
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5. **Page 1 — Home** — full content inventory
6. **Page 2 — Solutions** — full content inventory
7. **Page 3 — Sectors** — full content inventory
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11. **Technical implementation** — stack, API, accessibility, performance
12. **Placeholders & pre-launch checklist**

1 • Overview

The Aeronive Labs website is a four-page marketing site presenting a compliance-native AI consultancy serving regulated sectors. It is built as a statically generated Next.js application with a real-time 3D hero rendered in WebGL.

The central argument the site makes, repeated across every page: *frontier AI capability is only useful to a regulated organisation if it can run inside their perimeter and explain itself to an auditor*. Every section supports that claim rather than listing features.

At a glance

Pages	Home, Solutions, Sectors, Contact
API routes	POST /api/contact
Rendering	Static prerender for all 4 pages; contact API server-rendered on demand
Framework	Next.js 16.2 (App Router), React 19.2, TypeScript
Styling	Tailwind CSS v4 with a custom token layer
3D	React Three Fiber 9 + three.js 0.185, three custom GLSL shaders
Animation	Motion 12 (scroll reveals, parallax), Lenis 1.3 (momentum scroll)
Theme	Dark only — near-black base #04050a with a single blue accent
Source files	~3,650 lines across 23 components, 2 lib modules, 4 pages

Content model

Almost all copy is centralised in a single module, `src/lib/site.ts`, so text can be edited without touching layout. Longer-form page prose (the Solutions deep-dives, the Sectors framework matrix) sits in a data array at the top of its own page file.

Exported from <code>site.ts</code>	Used by
site	Metadata, footer, contact page, contact form fallback
nav	Header navigation, mobile sheet
frameworks	Home marquee, footer strip
chips	Home capability rail beneath the console
capabilities	Home bento grid
metrics	Home metrics strip
process	Home process section, Contact "what happens next"
sectors	Home sector grid, Sectors page, footer, contact form dropdown
topologies	Solutions deployment section
faqs	Contact page FAQ

2 · Brand & positioning

Name	Aeronive Labs
Tagline	Compliance-native AI for frontier systems
Domain	aeronivelabs.com (placeholder — set in <code>site.url</code>)
Contact	contact@aeronivelabs.com (placeholder — set in <code>site.email</code>)

Positioning statement

`SITE.DESCRPTION` — USED IN METADATA AND THE HERO

Aeronive Labs builds compliance-native AI systems for regulated sectors — private Company Brains, local-first interconnects, and data-backed pipelines that run inside your perimeter.

The logo mark

A sealed hexagonal boundary containing a three-node lattice. The boundary reads as a perimeter or seal; the lattice inside is the same knowledge-graph motif the 3D hero animates. Three nodes are coloured across the brand gradient (light blue → blue → cyan). Rendered as inline SVG at `src/components/ui/Icons.tsx`, so it stays crisp at any size and needs no image asset.

The wordmark pairs the mark with "Aeronive" in near-white and "Labs" in muted grey, so the company name dominates.

Voice

The copy throughout is deliberately plain and slightly adversarial toward industry norms. It avoids superlatives and vendor language, and repeatedly names the failure mode it is arguing against. Recurring devices:

- **Negation as a headline.** "Less pilot theatre." · "Four phases, no theatre." · "Compliance is not a horizontal layer." · "Defensible under scrutiny, not just compliant on paper."
- **Concrete over abstract.** "a refinery, a vessel, a field hospital" rather than "challenging environments".
- **Naming the alternative.** "Guardrails written as prompt text are not controls — they are suggestions."
- **Understatement in commitments.** "We do not grade our own homework." · "No slideware."

Recurring vocabulary

Terms used consistently across pages and worth keeping stable in future copy:

Company Brain

local-first interconnect

data-backed pipelines

perimeter

egress

lineage

regime

risk-tiered

deterministic replay

air-gapped

evidence

assurance

retrieval

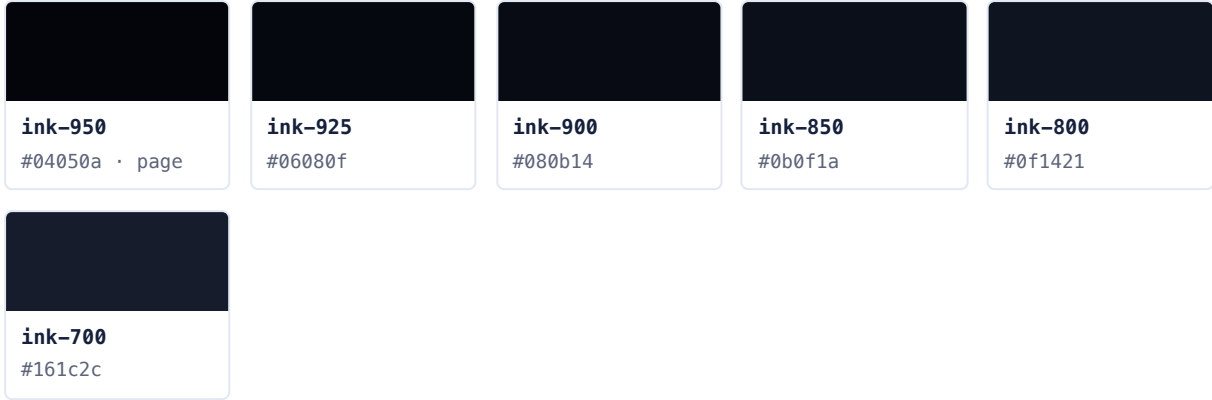
citations

3 · Design system

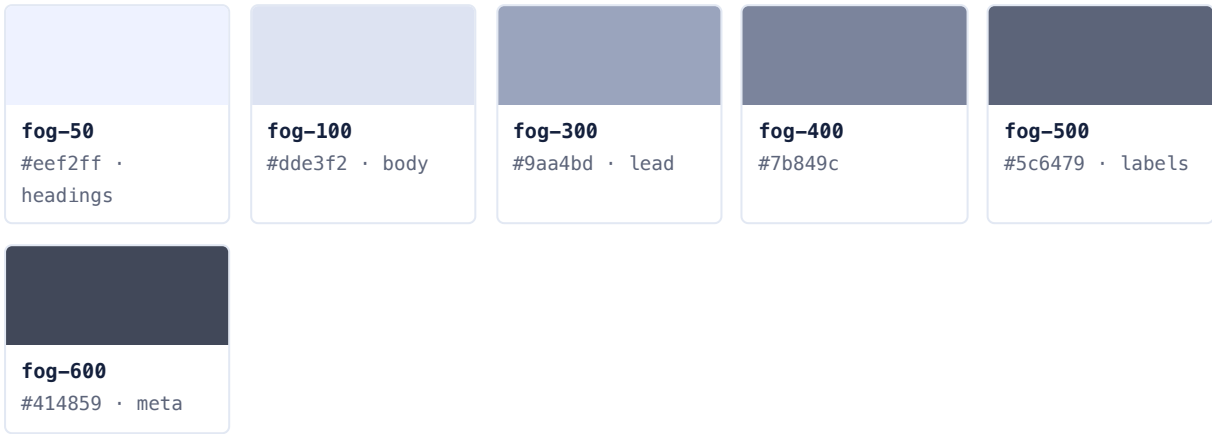
All tokens are defined in `src/app/globals.css` inside Tailwind v4's `@theme` block, which makes them available as utilities (`bg-ink-950` , `text-fog-300` , and so on).

Surfaces — "ink"

A near-black base, shifted slightly cool so it reads as deep blue-black rather than grey.

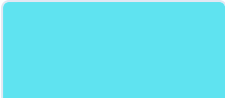
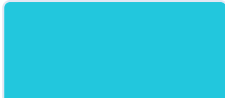


Text — "fog"



Accent — "beam", plus secondaries



 aqua-400 #5fe3f0 · live dot	 aqua-500 #22c7dd	 violet-400 #a78bfa · reserved
---	---	--

Status colours borrow from Tailwind defaults: emerald for pass/verified states, amber for degraded or flagged states, red for form validation errors.

Typography

Role	Face	Treatment
Display & body	Geist Sans (variable, self-hosted via next/font)	Headings at weight 500, letter-spacing -0.03em, text-wrap: balance
Labels & data	Geist Mono	Uppercase, letter-spacing 0.14-0.18em, 11px, muted

Headings scale fluidly with clamp() rather than at breakpoints. The hero is clamp(2.2rem, 7.2vw, 5.4rem) ; section headings are clamp(2rem, 4.4vw, 3.4rem) . Two gradient text treatments are used: .text-gradient (white → periwinkle → slate, for most headings) and .text-gradient-beam (blue → cyan → blue, reserved for the emphasised half of a hero headline).

Surface treatments

- .card — 18px radius, 7%-white hairline border, a top-down white gradient wash over a translucent ink fill, and a 12px backdrop blur. A ::before pseudo-element draws a 1px blue highlight along the top edge, which is what makes dark cards read as lit rather than flat.
- .glass — heavier blur with 140% saturation, used for the floating nav pill and mobile controls.
- .grain — an inline SVG fractal-noise overlay at 16% opacity in overlay blend mode. Its job is killing gradient banding across the large dark fills.
- .grid-backdrop — a 64px engineering grid, radially masked so it fades out at the edges.
- .rule-fade — a 1px divider that fades to transparent at both ends.
- .eyebrow — the pill capsule above headings, with a pulsing cyan dot.

Motion

Behaviour	Implementation
Scroll reveal	<code>Reveal</code> — fade + 24px rise + 6px blur clear, 0.85s on a <code>[0.22, 1, 0.36, 1]</code> ease, fires once, staggered by index
Hero parallax	<code>ScrollParallax</code> — the console lifts and scales from 0.96 → 1 as the hero scrolls past
Momentum scroll	Lenis, lerp 0.11 — disabled on coarse pointers and for reduced-motion users
Marquee	42s linear translate of a duplicated list, edge-masked
Card hover	3px lift, border shifts to 24% beam, top highlight to full opacity, 0.45s
Live dot	2.4s opacity + expanding ring pulse

Reduced motion. A global media query collapses all animation and transition durations to ~0. Independently, the 3D canvas is not mounted at all, Lenis is bypassed, and `Reveal` / `ScrollParallax` render their children statically.

CSS layering rule. Every custom class lives inside `@layer components` (and base styles inside `@layer base`). Unlayered CSS outranks all layered Tailwind utilities, so a bare `.btn { display: inline-flex }` silently defeats `hidden` and `sm:inline-flex`. This caused a real responsive bug during build and must be preserved.

4 · Site architecture

Route	Title	Rendering
/	Aeronive Labs — Compliance-native AI for frontier systems	Static
/solutions	Solutions — Aeronive Labs	Static
/sectors	Sectors — Aeronive Labs	Static
/contact	Contact — Aeronive Labs	Static
/api/contact	—	Server, on demand

Header navigation

Fixed, 68px tall, transparent at the top of the page and transitioning to a blurred ink background with a hairline border once scrolled past 24px.

- **Left** — logo mark + "Aeronive Labs" wordmark, links home
- **Centre** — a floating glass pill containing Home · Solutions · Sectors · Contact; the active route gets a filled chip. Hidden below 768px.
- **Right** — "Book a briefing" primary button (hidden below 640px) and a hamburger toggle (shown below 768px)

The mobile sheet is a full-height blurred panel listing all four routes at 24px with numbered indices (01–04), a divider under each, and the primary CTA at the bottom. Opening it locks body scroll; tapping any link closes it.

Footer

Four columns beneath a radial blue horizon glow, then a bottom bar.

Column	Contents
Brand	Mark + wordmark; "Compliance-native AI systems for regulated sectors. Built to run inside your perimeter and stand up to an audit."; email link
Solutions	Company Brain · Local-first interconnect · Data-backed pipelines · Deployment topologies (deep links into /solutions)
Sectors	Financial services · Healthcare & life sciences · Public sector & defense · Legal & professional services (first four, deep-linked)
Company	Contact · Compliance posture · Engagement model
Bottom bar	© {current year} Aeronive Labs. All rights reserved. — plus EU AI Act · ISO/IEC 42001 · NIST AI RMF · SOC 2 Type II · HIPAA

5 · Page 1 — Home /

Eight sections top to bottom: hero, regime rail, capability bento, metrics, sectors, process, CTA, footer.

5.1 Hero

EYEBROW PILL (FLANKED BY FADING TICK RULES)

● COMPLIANCE-NATIVE AI INFRASTRUCTURE

H1 — LINE 1 IN WHITE GRADIENT, LINE 2 IN BEAM GRADIENT

**Frontier AI,
governed by design.**

SUB-HEADLINE

Aeronive Labs builds compliance-native AI for regulated sectors — private Company Brains, local-first interconnects, and data-backed pipelines that run inside your perimeter and hold up under audit.

Buttons: "Book a technical briefing" (primary → /contact) · "Explore the platform" (ghost → /solutions)

Caption beneath: ON-PREMISE · PRIVATE VPC · AIR-GAPPED

Behind the copy: the full 3D hero scene, an engineering grid, and a radial scrim that guarantees headline contrast over the brightest part of the light rift (wider and stronger below 640px).

The Company Brain console

A static product illustration built entirely in HTML and CSS — no screenshot, no image asset. It sits below the hero copy and lifts on scroll. Contents:

- **Title bar** — logo, company-brain / procurement , and two status chips: ● ON-PREM (green) and 🔒 NO EGRESS
- **Left rail** — seven icons (network, database, policy, layers, gauge, server, shield); the first is active
- **Query bar** — "Which vendors have unresolved SOC 2 exceptions past renewal?" with a ⚠️ hint
- **Grounded answer** — "Four vendors carry unresolved Type II exceptions past their renewal date.¹ Two are classified **material** under the procurement risk policy and require CISO sign-off before auto-renewal.² The remaining two fall inside the 30-day remediation window.³" — with a blinking cursor and three inline citation chips
- **Policy checks** — Access: Enforced · PII redaction: 3 spans · Jurisdiction: EU-West · Egress: Blocked (each with a green tick)
- **Eval strip** — "Eval · retrieval precision — 97.2%" over a 12-bar chart, last three bars highlighted in blue
- **Retrieval trace panel** — Vendor_MSA_2024.pdf (§4.2 · p.17, 96) · SOC2_Type_II_Q3.pdf (Exceptions · p.4, 91) · risk_register.sql (vendor_findings, 84) · Procurement wiki (Escalation policy, 72), each with a confidence bar
- **Audit record** — trace_id 8f2c·a91e · actor jdoe@corp · scope procurement:read · signed ✓

Capability rail

Six items in a hairline-separated grid directly beneath the console:

Zero data egress	Air-gap capable	Full audit trail
Model agnostic	Offline tolerant	Cited answers

5.2 Regime rail

Label: BUILT AGAINST THE REGIMES THAT BIND YOU , above an infinite edge-masked marquee of ten framework names:

EU AI Act

ISO/IEC 42001

NIST AI RMF

SOC 2 Type II

HIPAA

GDPR

DORA

ISO 27001

FedRAMP-aligned

PCI DSS

This replaces the customer-logo bar in the reference designs — it makes a claim about what the systems are built against rather than implying clients we cannot name.

5.3 Capability bento

SECTION HEADING

Less pilot theatre.

Systems that survive contact.

Six building blocks we compose into a system shaped by your data estate, your regulator, and your network reality — not by a reference architecture.

A 12-column asymmetric grid: 7+5, then 4+4+4, then a full-width card. Each card carries a bespoke schematic diagram.

Company Brain — 7 columns

Blurb. A private, governed model of everything your organization knows — documents, systems, decisions, and the tribal knowledge that never got written down.

Detail. Ingests from your existing stores behind the firewall. Every answer carries citations back to source, and access control is enforced at retrieval time rather than bolted on after generation.

Diagram. Five heterogeneous source documents converging along curved paths into a single glowing core node, which outputs one cited answer bar with three citation chips.

Local-first interconnect — 5 columns

Blurb. Custom services that keep working when the network doesn't.

Detail. Edge-resident inference, CRDT-backed sync, and deterministic replay so a site, vessel, or facility stays operational through disconnection and reconciles cleanly on reconnect.

Diagram. SITE A (serving) and SITE B (local only) either side of a dashed link labelled LINK DOWN, above a CRDT queue of pending ops — nine filled, five empty — and the note "deterministic replay on reconnect".

Data-backed pipelines — 4 columns

Blurb. Curation, labeling, lineage, and evaluation sets built from your own data.

Detail. Versioned and attributable end to end. If a regulator asks why the system produced an output, the pipeline can answer with the exact records involved.

Diagram. A four-stage chain — Source → Curate → Label → Eval — with tags `lineage:full`, `sha256` `pinned`, `v4` · `reproducible`.

Policy & guardrail engine — 4 columns

Blurb. Machine-readable policy compiled into runtime controls.

Detail. Redaction, jurisdiction routing, refusal boundaries, and escalation paths expressed as versioned artifacts — reviewable by counsel, enforced by the runtime.

Diagram. A `policy.v12.yaml` panel marked "signed", listing jurisdiction: eu-west-1 · pii.redact: [name, dob] · refuse_if: tier == high · escalate_to: human_review, each ticked.

Evaluation & assurance — 4 columns

Blurb. Continuous red-teaming, drift detection, and regulator-ready evidence generated from production traffic.

Detail. Evaluation is a standing process, not a launch gate. Evidence accumulates as the system runs.

Diagram. A ten-bar drift monitor over 30 days with one amber bar flagged, captioned "1 flagged → re-eval".

Sovereign deployment — full width

Blurb. On-premise, private VPC, or fully air-gapped. Your weights, your keys, your logs, no egress by default.

Detail. Deployment topology is a first-class design input, decided before the first line of integration code.

Diagram. Three topology tiles — Private VPC (your keys) · On-premise (your hardware) · Air-gapped (no network path).

5.4 Metrics strip

Figure	Caption
0 bytes	Client data leaving the perimeter under default deployment policy.
100 %	Generated answers traceable to a cited source record.
6 wks	Typical path from data assessment to a production pilot.
24/7	Continuous evaluation and drift monitoring once live.

These are positioning placeholders, not measured figures. Replace them with numbers you can substantiate before launch.

5.5 Sectors preview

SECTION HEADING – LEFT ALIGNED, WITH AN "ALL SECTORS" BUTTON OPPOSITE

**Regulated sectors,
on their own terms.**

Compliance is not a horizontal layer. Each sector carries its own regime, its own evidentiary standard, and its own tolerance for failure.

Six cards in a three-column grid. Each shows the sector name, its summary, an arrow that slides on hover, and its regime chips. The whole card links to the matching anchor on /sectors. Content is listed in full in chapter 7.

5.6 Process

SECTION HEADING

Four phases, no theatre.

We scope against your regulatory surface before we write integration code, and we hand over something your team can defend without us in the room.

Four numbered steps across a faint connecting rail:

Step	Title	Body
01	Map	We assess the data estate, the regulatory surface, and how your use cases classify under the frameworks that bind you. The output is a risk-tiered scope, not a proposal deck.
02	Ground	We build the Company Brain: ingestion, lineage, the access model, and an evaluation set drawn from your real queries. Nothing ships against generic benchmarks.
03	Interconnect	Local-first services are wired into existing systems — designed to degrade gracefully, tolerate disconnection, and reconcile deterministically.
04	Assure	Continuous evaluation, audit evidence, and change control. Your team takes operational ownership with the runbooks to keep it defensible.

5.7 Closing CTA

CTA PANEL — GRAIN, GRID BACKDROP, BLUE GLOW RISING FROM BELOW

Bring frontier capability inside the perimeter.

A technical briefing covers your data estate, the regimes that apply, and what a first production use case would actually take. No slideware.

Buttons: "Book a technical briefing" → /contact · "Read the platform detail" → /solutions

6 · Page 2 — Solutions /solutions

Meta description: "Company Brain, local-first interconnects, and data-backed pipelines — compliance-native AI systems deployed on-premise, in a private VPC, or fully air-gapped."

6.1 Page hero

EYEBROW · H1 · LEAD

● PLATFORM

One system, assembled from four disciplines.

We do not sell a product with a compliance module bolted on. We build the retrieval, interconnect, data, and governance layers as one system, shaped by the regime you operate under.

6.2 Deep-dive blocks

Four alternating two-column blocks, each pairing prose and a checklist with a framed diagram. Anchors: `#company-brain`, `#local-first`, `#data-pipelines`, `#governance`.

01 · Company Brain

Everything your organization knows, made retrievable and governed.

Most enterprise knowledge is not in a database. It is spread across contract archives, ticket histories, engineering wikis, regulatory filings, and the judgement of people who have been there fifteen years. A Company Brain consolidates that into a single governed index that sits behind your firewall.

Retrieval is permission-aware before generation, not after. A user who cannot open a document in the source system cannot cause it to influence an answer — the access model is evaluated at query time against your existing identity provider.

- Ingestion from document stores, databases, ticketing, and code
- Citations resolve to the exact span in the source record
- Access enforced at retrieval against your existing IdP
- Incremental re-indexing as source systems change

02 · Local-first interconnect

Custom services that keep working when the network doesn't.

A refinery, a vessel, a field hospital, and a classified facility have one thing in common: the connection is not guaranteed. Systems built on the assumption of a live link to a hosted model fail exactly when they are needed most.

We build interconnecting services that treat disconnection as the normal case. Inference runs on local hardware, state is held in conflict-free replicated types, and reconciliation on reconnect is deterministic and replayable — so an auditor can reconstruct what the system knew at any point in time.

- Edge-resident inference on local hardware
- CRDT-backed state, no lost writes during partition
- Deterministic replay for audit reconstruction
- Signed, offline update bundles for air-gapped sites

03 · Data-backed pipelines

Curation, lineage, and evaluation built from your own data.

Generic benchmarks tell you nothing about whether a system works on your corpus. We build evaluation sets from your real queries and your real documents, graded by your subject-matter experts, and we hold releases against them.

Every artifact in the pipeline is versioned and content-addressed. When a regulator asks why the system produced a specific output on a specific date, the answer is a lookup rather than an investigation.

- Expert-graded evaluation sets drawn from real usage
- Content-addressed, reproducible pipeline artifacts
- Full lineage from source record to generated span
- Structured labeling workflows with reviewer agreement tracking

04 · Policy & assurance

Policy your counsel can read, enforced by the runtime.

Guardrails written as prompt text are not controls — they are suggestions. We compile machine-readable policy into runtime enforcement: redaction spans, jurisdiction routing, refusal boundaries, and escalation to human review.

Assurance runs continuously rather than as a launch gate. Drift detection, adversarial probing, and evidence generation operate against production traffic, so the audit file is a byproduct of running the system.

- Versioned, signed policy artifacts under change control
- Runtime redaction and jurisdiction-aware routing
- Continuous adversarial probing and drift detection
- Evidence generated from production, not assembled before an audit

6.3 Deployment topologies

SECTION HEADING

Where it runs is a design input, not an afterthought.

Topology is decided in the first week, because it constrains everything downstream — model choice, key management, update path, and what evidence the system can produce.

Topology	Posture	Description	Traits
Private VPC	Your cloud account	Runs inside your existing cloud boundary using your KMS keys, your VPC, your logging. We never hold credentials to production.	Customer-managed keys · Private networking · Existing IAM
On-premise	Your hardware	Deployed to your own datacenter or colocation. Open-weight models served locally, with no dependency on an external inference provider.	Local weights · No external inference · Hardware sizing included
Air-gapped	No network path	Fully disconnected operation with signed, offline update bundles and deterministic replay for audit reconstruction.	Offline update bundles · Deterministic replay · Physical media transfer

6.4 Integration surface

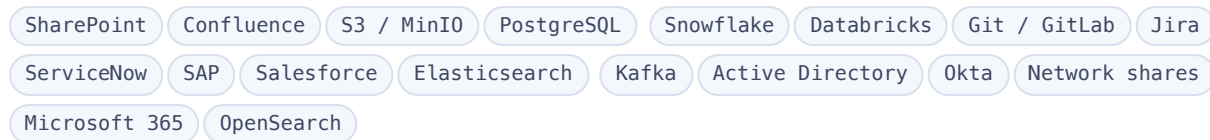
SECTION HEADING

Meets your estate where it already is.

Connectors run inside your network and authenticate as a service principal you control. Nothing is copied to infrastructure we operate.

Inline note: "Where a source system has no suitable API, we build the connector as part of the engagement and hand over the source."

Eighteen connector targets in a hairline grid:



6.5 Assurance strip

LABEL: CONTINUOUS ASSURANCE

Evidence accumulates while the system runs.

Drift against your evaluation set is monitored continuously. When a regression trips a threshold, the affected release is flagged and routed for re-evaluation before it reaches users.

6.6 Closing CTA

Start with the assessment.

Two weeks against your data estate and regulatory surface produces a risk-tiered scope and a concrete first use case.

Buttons: "Book a technical briefing" → /contact · "See sector detail" → /sectors

7 · Page 3 — Sectors /sectors

Meta description: "Compliance-native AI for financial services, healthcare, public sector, legal, energy, and insurance — mapped to the regimes that govern each."

7.1 Page hero

● SECTORS

Compliance is not a horizontal layer.

Each regulated sector carries its own regime, its own evidentiary standard, and its own definition of an unacceptable failure. We build against the specific one that binds you.

7.2 Sector detail

Six numbered rows separated by hairlines. Each pairs the sector name, summary and regime chips on the left with a "Representative use cases" card on the right. Anchors match the slugs below.

01 · Financial services #financial-services

Model risk management, surveillance, and client-facing systems where every recommendation has to be explainable to a supervisor.

Regimes: DORA · SR 11-7 · MiFID II · SOC 2

- Credit memo drafting with full source attribution
- Trade surveillance triage with reviewable rationale
- Model risk documentation kept current automatically

02 · Healthcare & life sciences #healthcare

Clinical and research workloads where PHI never leaves the estate and provenance is a regulatory requirement, not a feature.

Regimes: HIPAA · GxP · EU AI Act · ISO 13485

- Protocol and submission drafting against internal precedent
- Prior-authorization evidence assembly
- Safety signal triage across trial and literature corpora

03 • Public sector & defense #public-sector

Classified and controlled environments running fully disconnected, with deterministic behavior and complete audit reconstruction.

Regimes: FedRAMP-aligned · CMMC · NIST 800-53 · ITAR

- Air-gapped analyst assistance over controlled corpora
- Field-deployed inference tolerant of comms loss
- Records and FOIA response preparation

04 • Legal & professional services #legal

Privilege-aware systems where matter boundaries, conflicts, and confidentiality are enforced structurally.

Regimes: Privilege rules · GDPR · ISO 27001 · Client MSAs

- Matter-scoped research that cannot cross a conflict wall
- Diligence review with reviewable extraction lineage
- Precedent retrieval across the firm's own work product

05 • Energy & industrials #energy

Plants, grids, and remote sites where connectivity is intermittent and the control environment is unforgiving.

Regimes: NERC CIP · IEC 62443 · ISO 45001 · OSHA

- Local-first maintenance and procedure assistance
- Incident reconstruction from operational history
- Permit and compliance documentation generation

06 • Insurance #insurance

Underwriting and claims systems subject to fairness testing, adverse-action explanation, and state-level scrutiny.

Regimes: NAIC AI Model Bulletin · Solvency II · GDPR · SOC 2

- Submission intake and risk summarization
- Claims triage with documented decision rationale
- Bias testing across protected-class proxies

7.3 Compliance posture #frameworks

What each regime actually requires of the system.

Not a badge wall. These are the obligations we design the architecture around, and what the running system produces to satisfy them.

Framework	Scope	What we do about it
EU AI Act	Risk classification & technical documentation	Use cases are classified by risk tier during scoping. High-risk deployments get the logging, human-oversight, and technical documentation the tier requires, generated by the running system.
ISO/IEC 42001	AI management system	We structure the engagement so your AI management system has real artifacts behind it — defined roles, impact assessments, and a change-control record for every model and policy version.
NIST AI RMF	Govern · Map · Measure · Manage	Our four-phase engagement maps directly onto the RMF functions, so the assessment output slots into an existing risk programme rather than sitting beside it.
SOC 2 Type II	Control evidence	Access enforcement, change management, and monitoring all emit evidence continuously. Control testing draws from the same audit log the system writes in normal operation.
HIPAA	PHI handling	PHI never leaves the covered entity's environment. Minimum-necessary access is enforced at retrieval, and every disclosure path is logged against an identified actor.
GDPR	Lawful basis & data subject rights	Lineage makes erasure and access requests tractable: because every generated span traces to source records, we can answer what was processed and remove it at source.

Closing line: "Operating under a regime not listed here? [Tell us which one.](#)" — links to /contact.

7.4 Assurance panel

Defensible under scrutiny, not just compliant on paper.

The difference between a system that passes review and one that survives an incident is whether it can explain itself afterwards.

Reconstructable decisions	Every output resolves to the retrieval set, policy version, model version, and actor identity that produced it.
Signed change control	Model, prompt, and policy changes are versioned artifacts with signed approval — no silent updates to a production system.
Human oversight paths	Escalation to a qualified reviewer is a runtime control with a recorded outcome, not a documented intention.
Independent evaluation	Evaluation sets are owned by your subject-matter experts. We do not grade our own homework.

7.5 Cross-link & CTA

A full-width link card: label "NEXT", heading "How the system is actually built" → /solutions.

Scope it against your regime.

Tell us the sector and the frameworks you answer to. We will come back with a risk-tiered view of which use cases are viable and what each one requires.

Buttons: "Book a technical briefing" → /contact · "Platform detail" → /solutions

8 · Page 4 — Contact /contact

Meta description: "Book a technical briefing with Aeronive Labs — scoped against your data estate, your sector, and the regimes you answer to."

8.1 Page hero

● CONTACT

Book a technical briefing.

Tell us what you are trying to build and which regime governs it. We will come back with a view on feasibility, the deployment topology it implies, and what a first production use case would take.

8.2 The form

Field	Type	Required	Placeholder / options
Name	text	Yes	Jordan Ellis
Work email	email	Yes	jordan@organization.com
Organization	text	No	Organization name
Sector	select	No	The six sectors, plus "Other"
Preferred deployment	radio pills	Defaulted	Private VPC · On-premise · Air-gapped · Not decided yet (default)
What are you trying to build?	textarea	Yes	"The systems involved, the regime you operate under, and what a successful first use case would look like."
website	hidden honeypot	—	Must stay empty; a filled value is silently accepted and discarded

Footnote beside the submit button: "We do not request access to production data at this stage." **Button:** "Request a briefing" → "Sending..." while in flight.

Success state

Request received.

We read every briefing request ourselves. Expect a reply within two business days, usually with a few questions about your data estate before we schedule anything.

Followed by a "Send another" button that resets the form.

Unconfigured / error state

An amber panel: "This form has no delivery endpoint configured yet, so nothing was sent. Email us directly and we will pick it up:" with an "Open email draft" button that opens a `mailto:` pre-populated with every field the user filled in.

8.3 Sidebar

Direct	contact@aeronivelabs.com — "For procurement, security questionnaires, or an existing engagement, email is usually faster."
No production access	Assessment runs against schemas, samples, and documentation you control.
NDA before detail	We sign yours. Technical discussion of your estate happens under it.
Engineers, not sales	The first call is with the people who would build the system.
What happens next	The four process steps (Map · Ground · Interconnect · Assure), each truncated to its first sentence.

8.4 FAQ

Heading: "Asked before every engagement." Four native `<details>` accordions with a plus icon that rotates to a minus on open.

Question	Answer
Do you require access to our production data?	No. Assessment runs against schemas, samples, and documentation you control. In every deployment topology we support, the system runs inside your perimeter and we hold no standing credentials to production.
Which models do you use?	We are model agnostic. Frontier hosted models where your policy allows it, open-weight models served locally where it does not. The retrieval, policy, and evaluation layers are designed so the model underneath can be swapped without re-architecting.
How do you handle the EU AI Act?	We classify each use case by risk tier during the mapping phase, then build the technical documentation, logging, and human-oversight controls that the tier requires. The evidence is generated by the running system rather than assembled by hand before an audit.
What does a first engagement look like?	A scoped assessment covering the data estate and regulatory surface, followed by a production pilot on one high-value use case. Most clients reach a working pilot inside six weeks.

9 • The 3D layer

Three shader-driven components composed into two scene variants. Everything is written as raw GLSL against custom `ShaderMaterial`s — there is no post-processing dependency, which keeps the bundle small and the frame cost predictable.

9.1 AuroraRift — the light rift

A single additive plane behind everything, driven by fractal Brownian motion noise. It produces two things at once: a narrow vertical rift down the centre of the composition, and two slow parabolic aurora bands sweeping across it.

- The rift is an exponential falloff from the centre line — a tight bright core inside a wider halo — with a noise-driven horizontal wobble so it never looks like a straight gradient.
- Vertical brightness is weighted toward the top of the frame, so the strongest part of the beam clears the headline rather than sitting behind it.
- Colour ramps from deep blue through the beam accent to near-white cyan at the hottest point.

9.2 Starfield — depth and parallax

1,600 points (900 on inner pages) distributed through a spherical shell, flattened vertically so the field reads as a horizon rather than a sphere. Each point carries its own seed and size attribute.

- Per-point twinkle on an offset sine, so no two stars pulse together.
- Roughly 6% render noticeably larger and brighter; about 28% run cool blue rather than near-white.
- The whole field rotates slowly and eases toward the pointer, giving parallax against the fixed copy.
- Positions come from a seeded PRNG, so the field is identical on every render.

9.3 NeuralCore — the Company Brain motif

The signature object, and the same idea as the logo mark rendered in three dimensions.

- **Lattice.** 200 nodes (140 on narrow screens) placed on a Fibonacci sphere for even coverage without pole clustering, then connected to their three nearest neighbours with duplicates removed — roughly 400 unique edges.
- **Signals.** Each edge carries a pulse that travels from one end to the other, wrapping continuously. Only a subset is live at any moment, so the lattice reads as a system under load rather than a decoration.
- **Nodes.** Points with a soft core and halo, breathing on independent phases; about 12% render as larger hub nodes.
- **Core.** An inverted-fresnel sphere at the centre — brightest facing the camera, with a rim falloff — kept deliberately dim because it sits behind the headline.
- **Rings.** Two counter-rotating additive tori, held close to the lattice radius. Wider rings sweep across the copy, which is why they are tight.
- The whole group rotates slowly and tilts toward the pointer.

9.4 Scene variants

	hero (/)	ambient (inner pages)
Rift intensity	1.05	0.5, wider and flatter
Stars	1,600, full parallax	900, half parallax
Lattice	Yes, behind the headline	No
Camera	z = 6, 45° FOV	z = 8, 45° FOV

9.5 Guards and fallbacks

SceneMount owns the decision of whether any 3D runs at all:

- Skipped entirely when the user prefers reduced motion.
- Skipped when the device reports no WebGL context (probed once and cached).
- Paused via `frameLoop="never"` when scrolled out of view, using an IntersectionObserver with 120px of margin.
- Device pixel ratio clamped to 1.75 so retina displays do not quadruple the fragment cost.
- The canvas is dynamically imported with SSR disabled, so three.js never blocks first paint.
- A CSS radial-gradient glow always renders underneath, so the hero looks intentional even with no canvas at all.

Two constraints when editing the shaders. Colours are hardcoded as sRGB literals because raw `ShaderMaterial` output bypasses three.js colour management. And `active` is a GLSL reserved word — it will not compile as a variable name.

10 · Component inventory

Component	Type	Role
Nav	client	Fixed header, centre pill nav, scroll state, mobile sheet with scroll lock
Footer	server	Four-column footer, horizon glow, framework strip, dynamic copyright year
SmoothScroll	client	Lenis wrapper; bypassed on coarse pointers and reduced motion
Hero	server	Home hero composition — scene, scrim, copy, console, capability rail
PageHero	server	Shared inner-page hero with the ambient scene
CTA	server	Closing panel; title, lead and both buttons are all props with defaults
ConsoleMockup	server	The Company Brain product illustration, pure HTML/CSS
FrameworkMarquee	server	Infinite regime rail; list duplicated once for a seamless –50% loop
ContactForm	client	Validation, submit lifecycle, success / unconfigured / error states, honeypot
CapabilityVisuals	server	Six schematic diagrams, reused at two sizes across Home and Solutions
Reveal	client	Scroll-triggered entrance; renders children plainly under reduced motion
ScrollParallax	client	Scroll-linked lift and scale for the hero console
Section	server	Container, Eyebrow, SectionHeading, Divider primitives
Icons	server	Logo, Wordmark and 14 stroke icons as inline SVG — no icon dependency
Scene / SceneMount	client	Canvas composition and the run/pause/skip decision
AuroraRift / Starfield / NeuralCore	client	The three shader pieces
client-env	lib	useMediaQuery, usePrefersReducedMotion, useCoarsePointer, useWebGL, useScrolledPast

11 • Technical implementation

11.1 Dependencies

Package	Version	Purpose
<code>next</code>	16.2.11	App Router, static generation, font optimisation
<code>react / react-dom</code>	19.2.4	—
<code>tailwindcss</code>	4.x	Utility layer and the <code>@theme</code> token system
<code>three</code>	0.185.1	WebGL renderer
<code>@react-three/fiber</code>	9.6.1	React reconciler for three.js
<code>@react-three/drei</code>	10.7.7	Helper library
<code>motion</code>	12.42.2	Scroll reveals and parallax
<code>lenis</code>	1.3.25	Momentum scrolling

11.2 The contact API

`POST /api/contact` — Node runtime.

1. Parses JSON; returns `400 invalid_json` on failure.
2. Checks the honeypot. A filled `website` field returns a success response without doing anything.
3. Coerces and length-caps every field (name 120, email 200, organization 160, message 5000).
4. Validates required fields and email shape; returns `422` with per-field errors that the form renders inline.
5. If `CONTACT_WEBHOOK_URL` is unset, logs a warning server-side and returns `503 unconfigured`.
6. Otherwise POSTs the payload as JSON with a source and timestamp; a non-2xx response returns `502 delivery_failed`.

The form does not deliver until `CONTACT_WEBHOOK_URL` is set. This is deliberate — the route reports `unconfigured` and the UI falls back to a prefilled email, rather than showing a success state for a message nobody received.

11.3 Accessibility

- Decorative canvases and glow layers are `aria-hidden` and non-interactive.
- Reduced motion is honoured at three levels: global CSS, the 3D mount, and every animation component.
- The mobile menu toggle carries `aria-expanded` and a state-dependent `aria-label`.

- Form fields use real `<label>` elements; errors render as text next to the field they belong to.
- The FAQ uses native `<details>` / `<summary>`, so it is keyboard-operable with no JavaScript.
- Focus styling on inputs is a visible ring, not a colour change alone.

11.4 Performance

- All four pages are prerendered as static HTML.
- `three.js` is dynamically imported and never blocks first paint; the hero has a CSS fallback beneath it.
- The render loop pauses entirely when the canvas leaves the viewport.
- Lattice geometry and starfield positions are computed once in `useMemo`.
- Fonts are self-hosted and subset by `next/font` with `display: swap`.
- Zero image assets — every graphic is SVG, CSS, or WebGL.

11.5 SEO metadata

- Title template `%s — Aeronive Labs`, with per-page titles and descriptions.
- Open Graph and Twitter card metadata on every page.
- `metadataBase` set from `site.url`; theme colour `#04050a`; colour scheme dark.
- Keywords covering compliance AI, frontier AI, Company Brain, local-first AI, EU AI Act, ISO 42001, air-gapped AI, regulated industries.

12 · Placeholders & pre-launch checklist

Everything below is content or configuration that was written to make the site complete and coherent, but which needs a decision or a real value before the site goes live.

Item	Where	Action
Metrics	<code>site.ts → metrics</code>	0 bytes · 100% · 6 wks · 24/7 are positioning claims, not measured figures. Replace with numbers you can substantiate, or reframe them as commitments.
Compliance claims	<code>/sectors</code>	The framework matrix describes a defensible architecture, but the specific regime statements should get a legal review.
Domain	<code>site.url</code>	Currently <code>https://aeronivelabs.com</code> . Drives <code>metadataBase</code> and all canonical URLs.
Email	<code>site.email</code>	Currently <code>contact@aeronivelabs.com</code> . Used in the footer, contact page, and the <code>mailto</code> fallback.
Form delivery	<code>CONTACT_WEBHOOK_URL</code>	Set in the deployment environment. Until then the form delivers nothing and says so.
OG image	<code>src/app/opengraph-image.png</code>	Metadata references a social preview image; no asset ships yet.
Console figures	<code>ConsoleMockup</code>	The 97.2% precision figure, vendor filenames and trace IDs are illustrative. They read as a product screenshot, so confirm you are comfortable with them.
Integration list	<code>/solutions</code>	Eighteen connector targets are listed as capability, not as shipped integrations. Trim to what you can actually deliver.
Favicon	<code>src/app/favicon.ico</code>	Still the Next.js default; replace with the Aeronive mark.

Commands

npm run dev	Development server on <code>:3000</code>
npm run build	Production build — currently compiles clean, 4 static pages
npm run lint	ESLint — currently passes with no errors

End of document · Aeronive Labs website content & design record · version 1.0